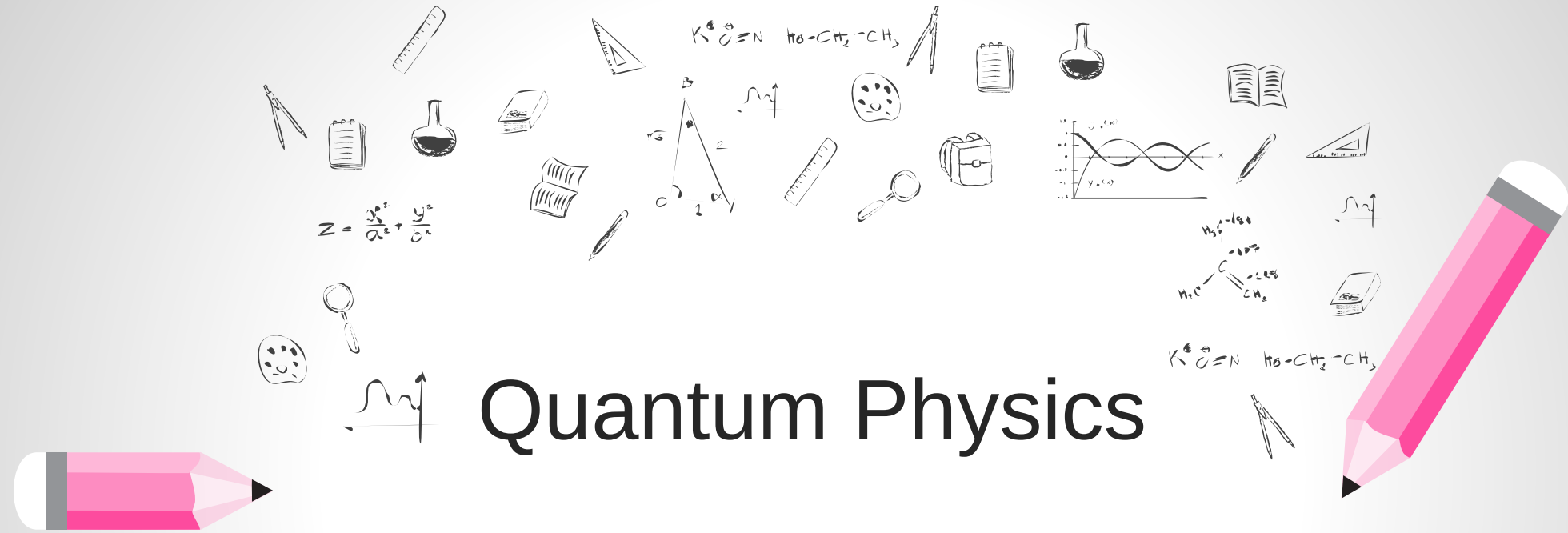




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Physics

Unit 1

01 Wave Particle Duality

02 Photoelectric Effect

03 Atomic Line Spectra

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Physics
Unit 2

Objectives

- ▶ understand how the behaviour of electromagnetic radiation can be described in terms of a wave model and a photon model, and how these models developed over time
- ▶ be able to use the equation $E = hf$, that relates the photon energy to the wave frequency
- ▶ understand that the absorption of a photon can result in the emission of a photoelectron
- ▶ understand how the photoelectric effect provides evidence for the particle nature of electromagnetic radiation
- ▶ understand how diffraction experiments provide evidence for the wave nature of electrons





Wave Particle Duality

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Wave properties of light

- ▶ Huygen's Principle – explains all the basic phenomena we see with light waves such as
 - ▶ reflection,
 - ▶ refraction,
 - ▶ diffraction and
 - ▶ interference



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Wave properties of light

- ⌚ Evidence that light is a wave
 - ⌚ Interference patterns produced by diffraction experiments
 - ⌚ Requires repeating cycles
 - ⌚ Polarization of light waves

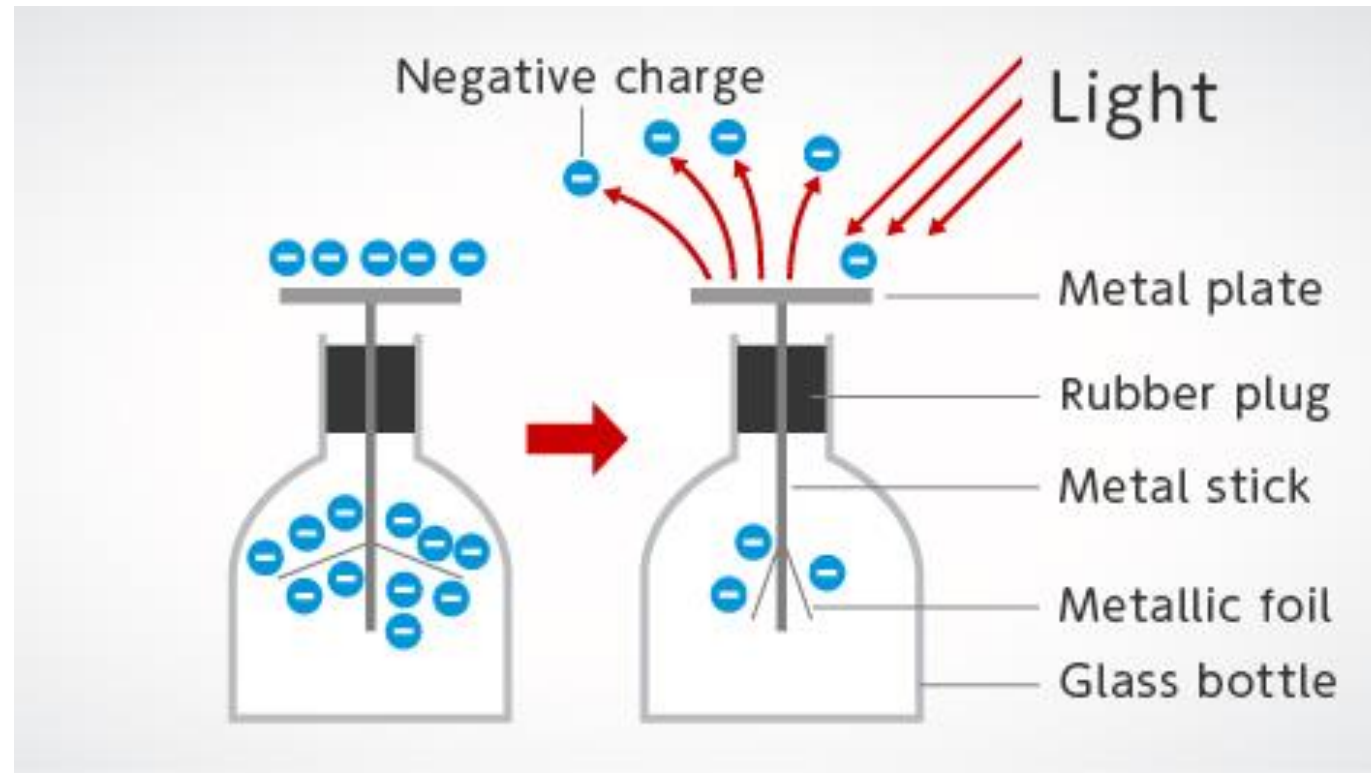


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Light as a particle

- ▶ Photoelectric Effect – evidence of particle nature
- ▶ One photon transfers energy to one electron and escapes the metal surface
- ▶ Charge reduces, so less repulsion. The metal foil falls

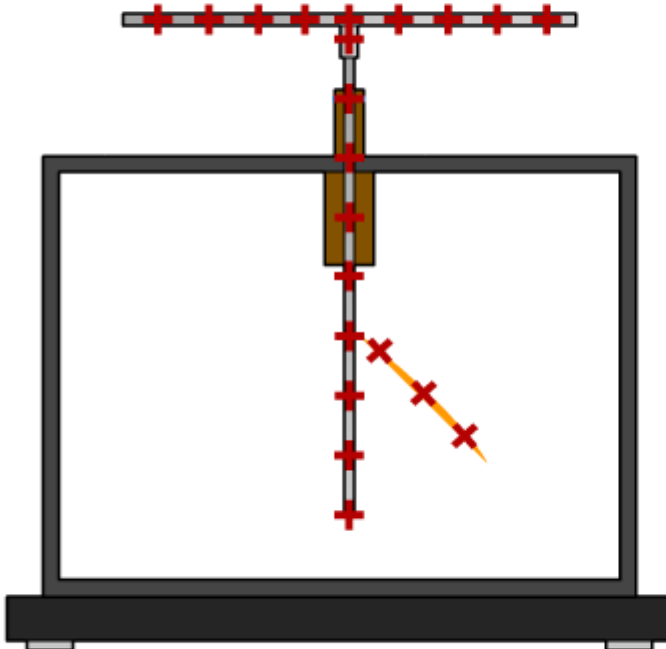


The Photoelectric effect



Photoelectricity

If the electroscope is **positively charged**, neither visible light or ultraviolet light has any effect on the gold leaf.



Navigation icons: trash can, highlighter, eraser, back, play, forward, refresh, and help.



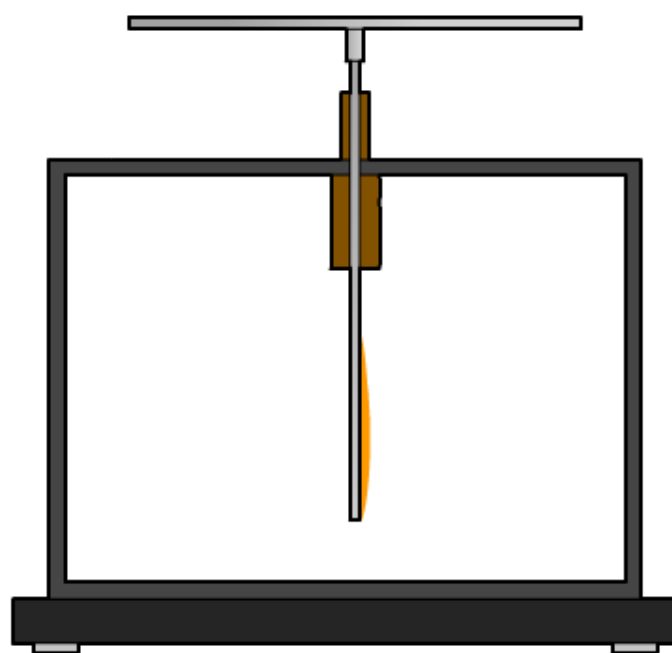
The Photoelectric effect



Photoelectricity

Photoelectricity can be 'seen' by using a **gold leaf electroscope**.

Click "**play**" or the electroscope to see what happens.



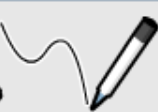
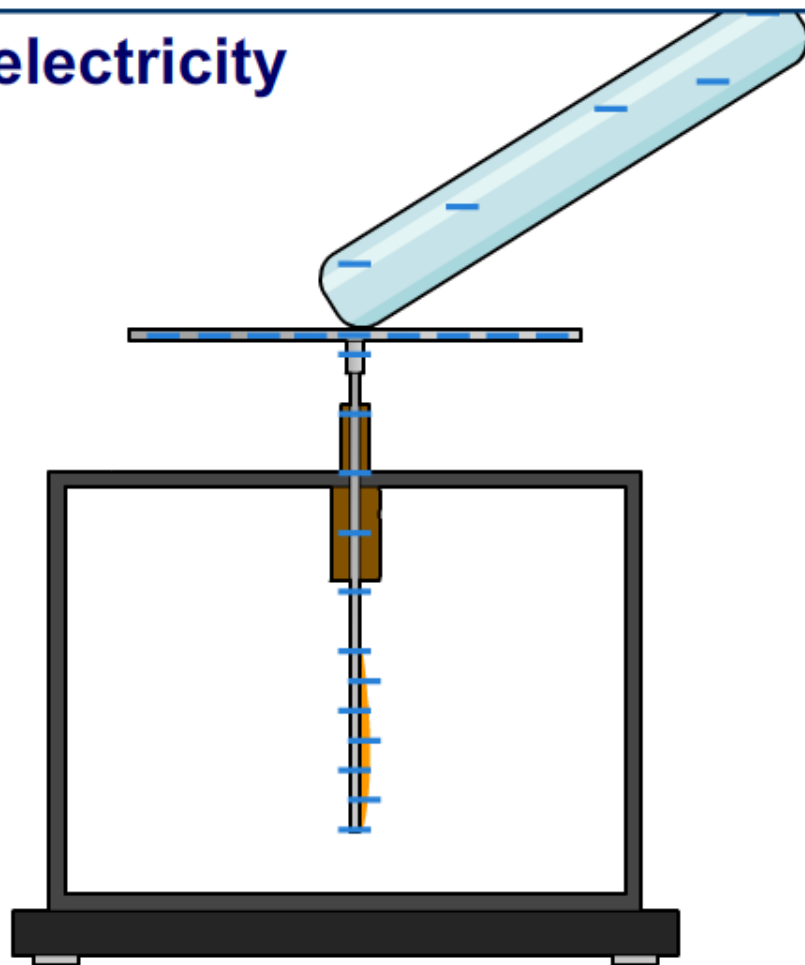
The Photoelectric effect



Photoelectricity

A piece of plastic is rubbed with a cloth so it becomes charged.

When the plastic is touched to the metal plate, the charge spreads down the rod and on to the gold leaf.



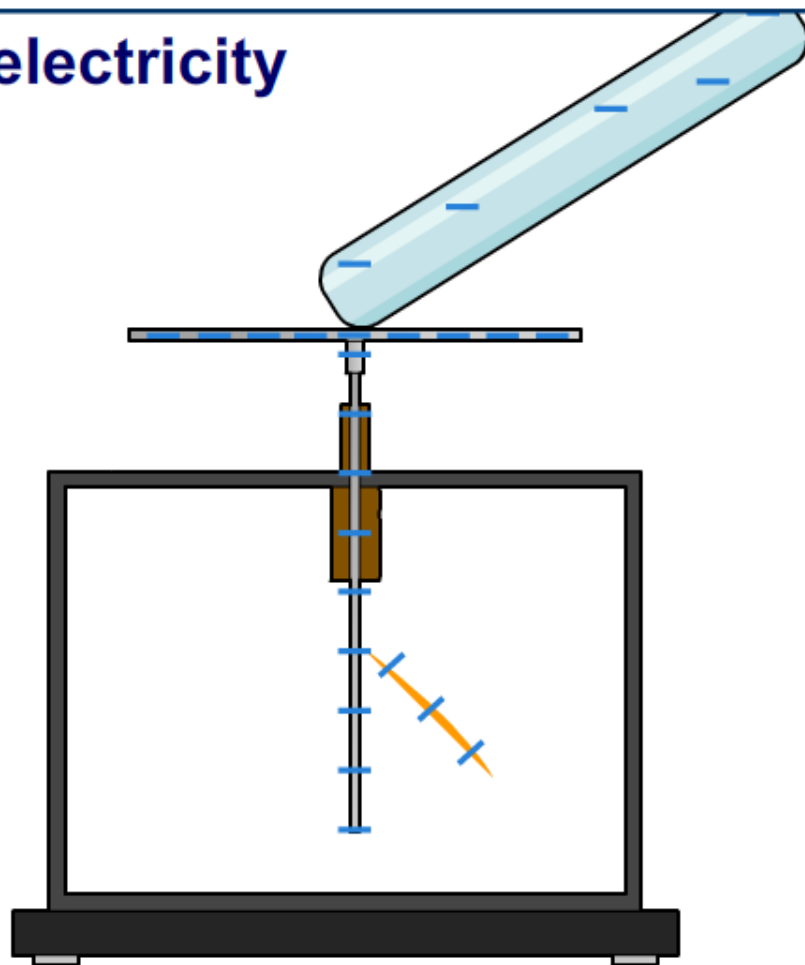
The Photoelectric effect



Photoelectricity

The metal rod and the gold leaf now both have a negative charge, so they repel each other.

The gold leaf therefore rises away from the metal rod.

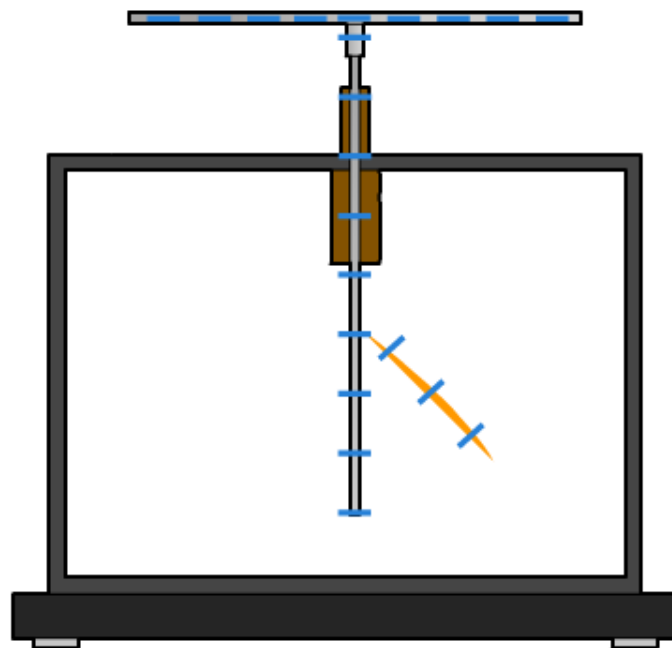


The Photoelectric effect



Photoelectricity

The metal rod and plate are isolated by an insulator, so when the plastic is withdrawn the negative charge cannot leak away. The gold leaf will stay up until the charge gradually disperses over time.



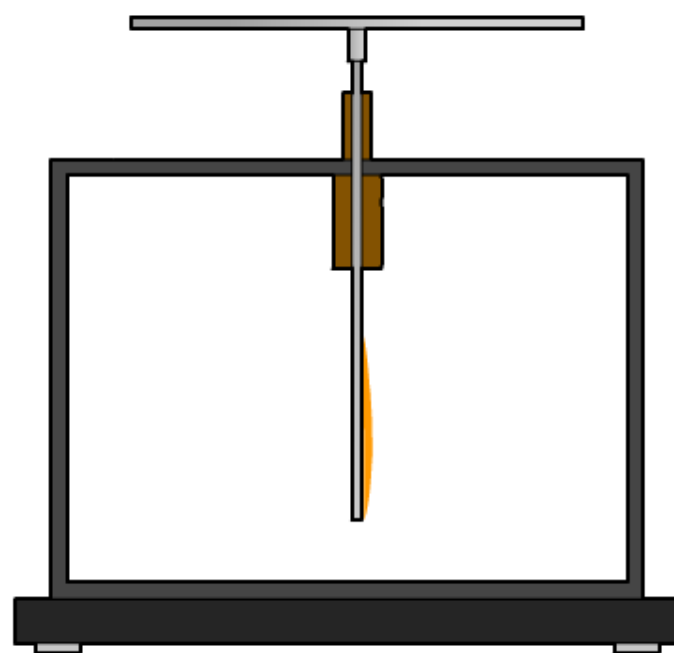
The Photoelectric effect



Photoelectricity



However, if **ultraviolet light** is shone on the plate, the electrons begin escaping from the top until all the charge is neutralized and the gold leaf drops.



The Photoelectric effect



Photoelectricity

If a sheet of glass is placed on top, the gold leaf stays up. This is because the ultraviolet light is absorbed by glass, and the electrons do not escape.

glass sheet

Navigation icons: back, forward, search, and a question mark.

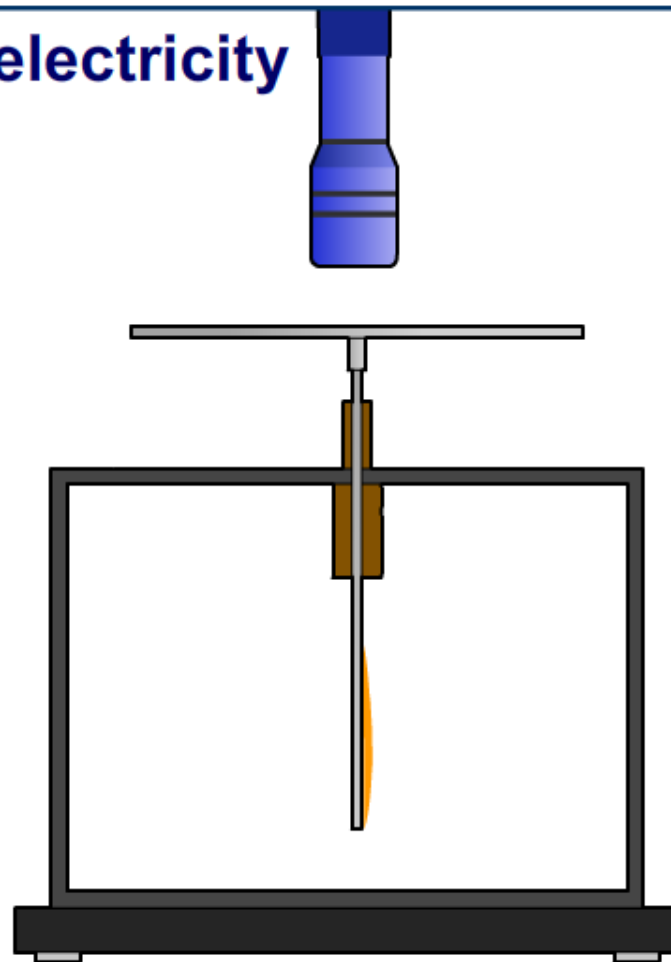


The Photoelectric effect



Photoelectricity

The rate of fall of the gold leaf depends on the distance of the light source from the metal plate. The further away the light source, the slower the rate; the closer the light source, the faster the rate.

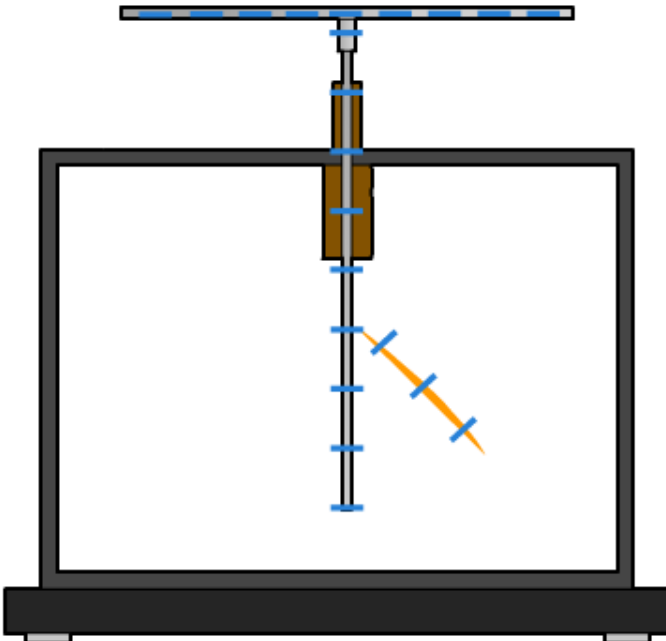


The Photoelectric effect



Photoelectricity

Visible light has no effect on the gold leaf.



Navigation icons: a trash can, a yellow highlighter, a black pen, a left arrow, a right arrow, a double right arrow, a refresh arrow, and a question mark.



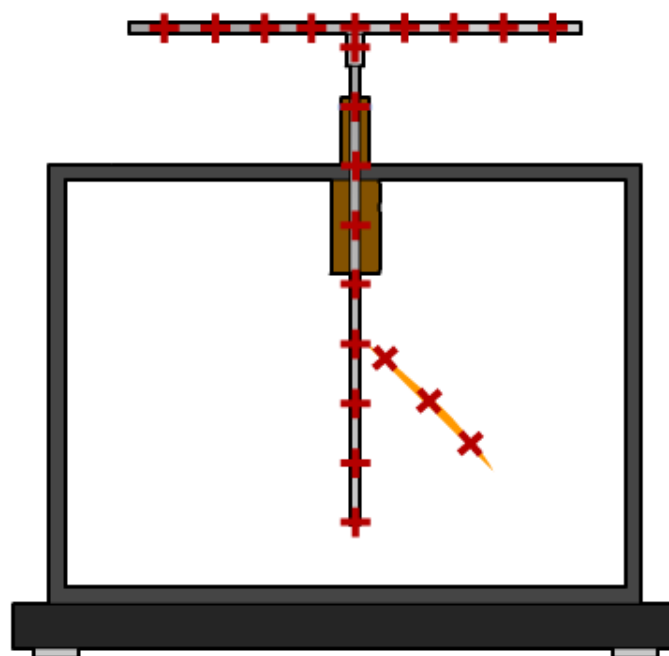
The Photoelectric effect



Photoelectricity



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Unit 2

How the Theory developed

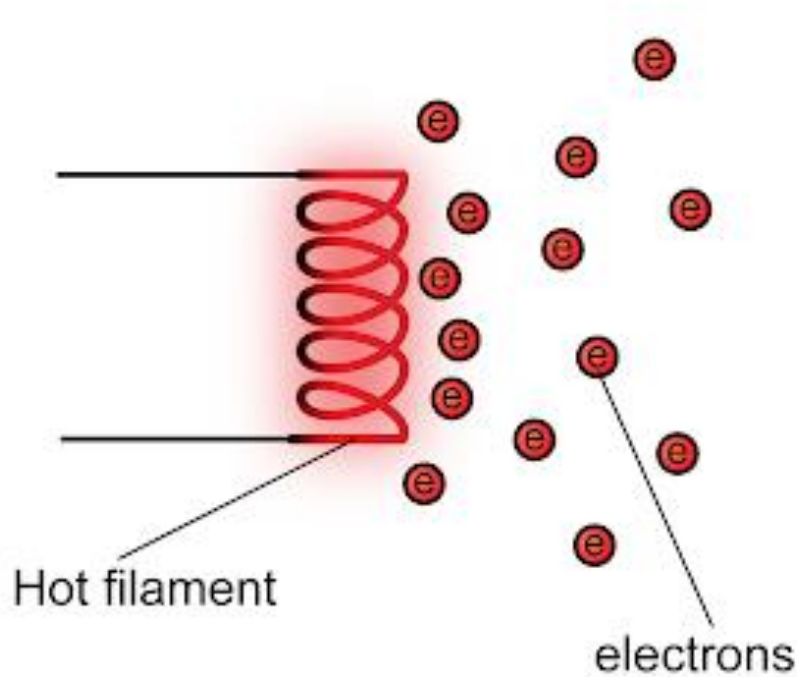
- ▶ First observed by Hertz in 1887
- ▶ The first explanation for the phenomenon was made in the beginning of the 20th century
- ▶ Planck in 1901 suggested light could exist as quantized particles called photons
- ▶ Einstein in 1905 explained the photoelectric effect
- ▶ The photoelectric effect cannot be explained using the wave theory of light



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Thermionic Emission



- ▶ Thermionic emission is a process of emission of charge particle (known as thermion) from the surface of a heated metal.
- ▶ Thermionic emission must NOT be confused with Photoelectric effect





Energy of a photon

$$E = h \times f$$

The relationship between the frequency of light and the energy of a single photon is shown in the equation above.

Press a variable in the equation at any point to rearrange the equation.

Press the forward and backward arrows to select an example questions, then press "=" to see the stages of working out.





Energy of a photon

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The Electronvolt, eV

A joule (J) is a large unit of energy when dealing with tiny atoms. Scientists often use an alternative unit for small amounts of energy, called an **electronvolt** (eV).

An electronvolt is equal to the amount of energy transferred to a single electron if it is accelerated through a potential difference of 1 V:

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$$

How many electronvolts to one joule?

$$1 \text{ J} = 1/(1.6 \times 10^{-19}) \text{ eV} = 6.25 \times 10^{18} \text{ eV}$$

Use these two conversion rates to change between the two. Be careful to use **joules** in calculations with other **SI units**.





Wave and photon calculations

Question: 1/4

What is the photon energy of electromagnetic radiation with a wavelength of $5.75 \times 10^{-12} \text{ m}$?

$$5.22 \times 10^{19} \text{ J}$$

$$1.14 \times 10^{-24} \text{ J}$$

$$1.27 \times 10^{-53} \text{ J}$$

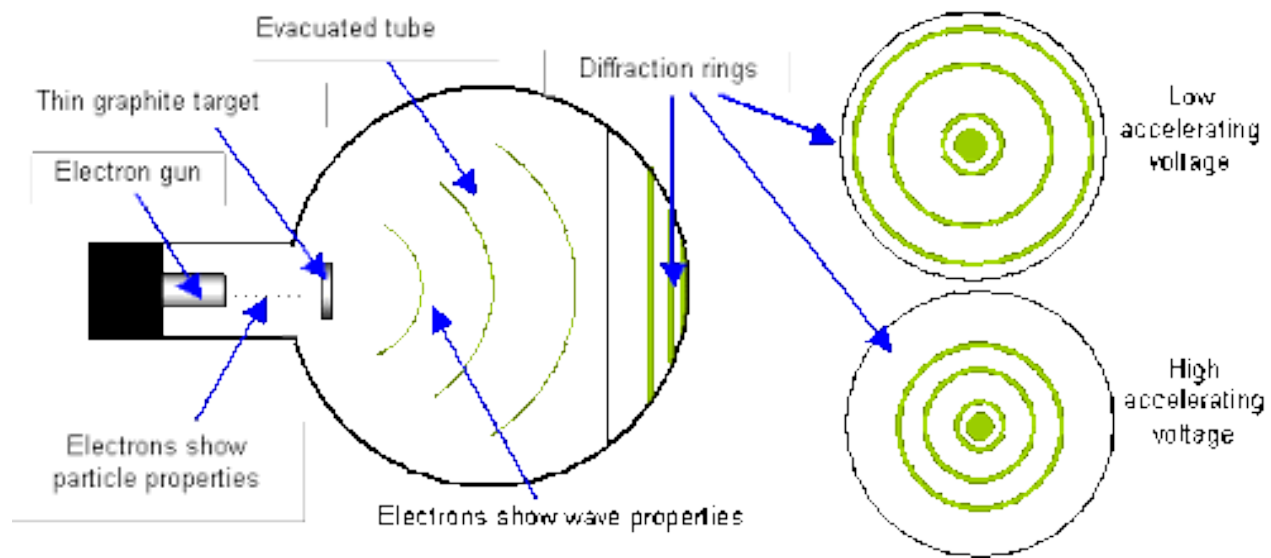
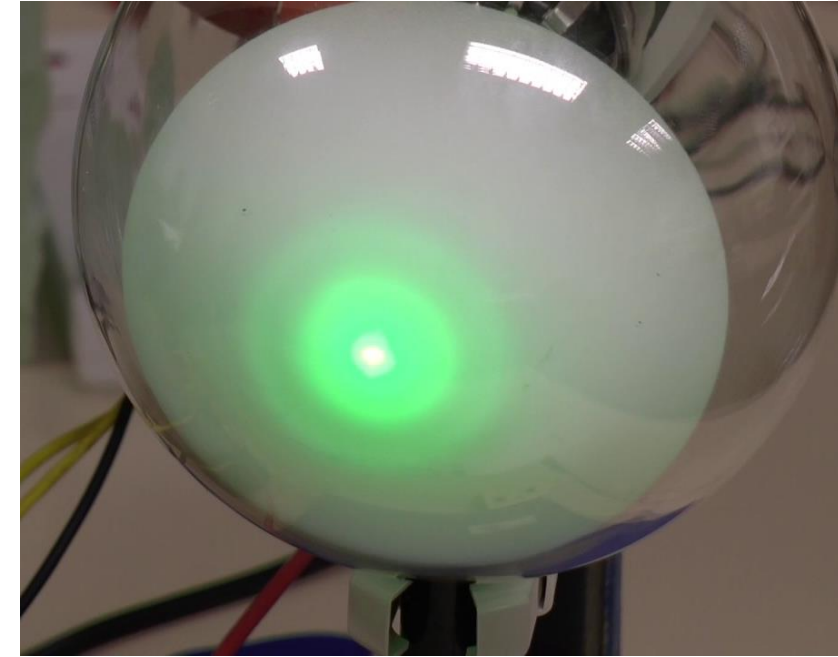
$$3.46 \times 10^{-14} \text{ J}$$



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Unit 2

Electrons as Waves



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Unit 2

Electrons as Particles

- ▶ To form an ion, an electron is lost – the atom loses a fixed lump of mass when it becomes an ion
- ▶ Mass to charge ratio is an identifying property of particles
- ▶ J. J. Thompson in 1897 demonstrated this property of electrons
- ▶ Millikan in 1913 found the charge of an electron
- ▶ The fact that electrons have a fixed amount of charge and a fixed mass indicates that they are localized particles



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Wave – Particle Duality

- EM waves and electrons behave as waves under some circumstances and behave as particles in others is known as wave particle duality

PHENOMENON	EVIDENCE FOR WAVES	EVIDENCE FOR PARTICLES
light	diffraction, interference, polarisation	photoelectric effect
electron	diffraction, interference	ionisation





Thank You

Insert the Sub Title
of Your Presentation